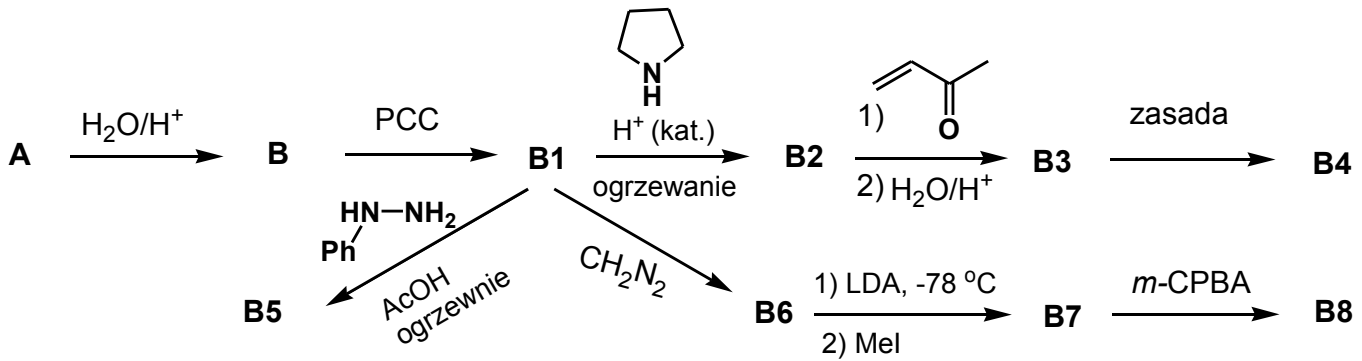
ZADANIE B7

Od węglowodoru do różnorodnych związków organicznych

Węglowodór **A** zawiera 87,8%$_{mas.}$ węgla, a w jego widmie ^{13}C NMR widoczne są trzy sygnały. Natomiast w widmie protonowym znajduje się sygnał przy 5,7 ppm oraz multiplety w zakresie 2,2-1,5 ppm o względnych integracjach 1:4. Węglowodór **A** może być wykorzystany do otrzymania szerokiej gamy istotnych związków organicznych, co przedstawiają poniższe schematy.

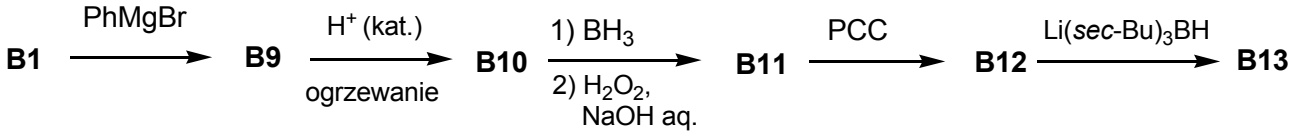


PCC - chlorochromian pirydyny; LDA - diizopropyloamidek litu; *m*-CPBA - kwas meta-chloroperoksybenzoesowy

Schemat 1

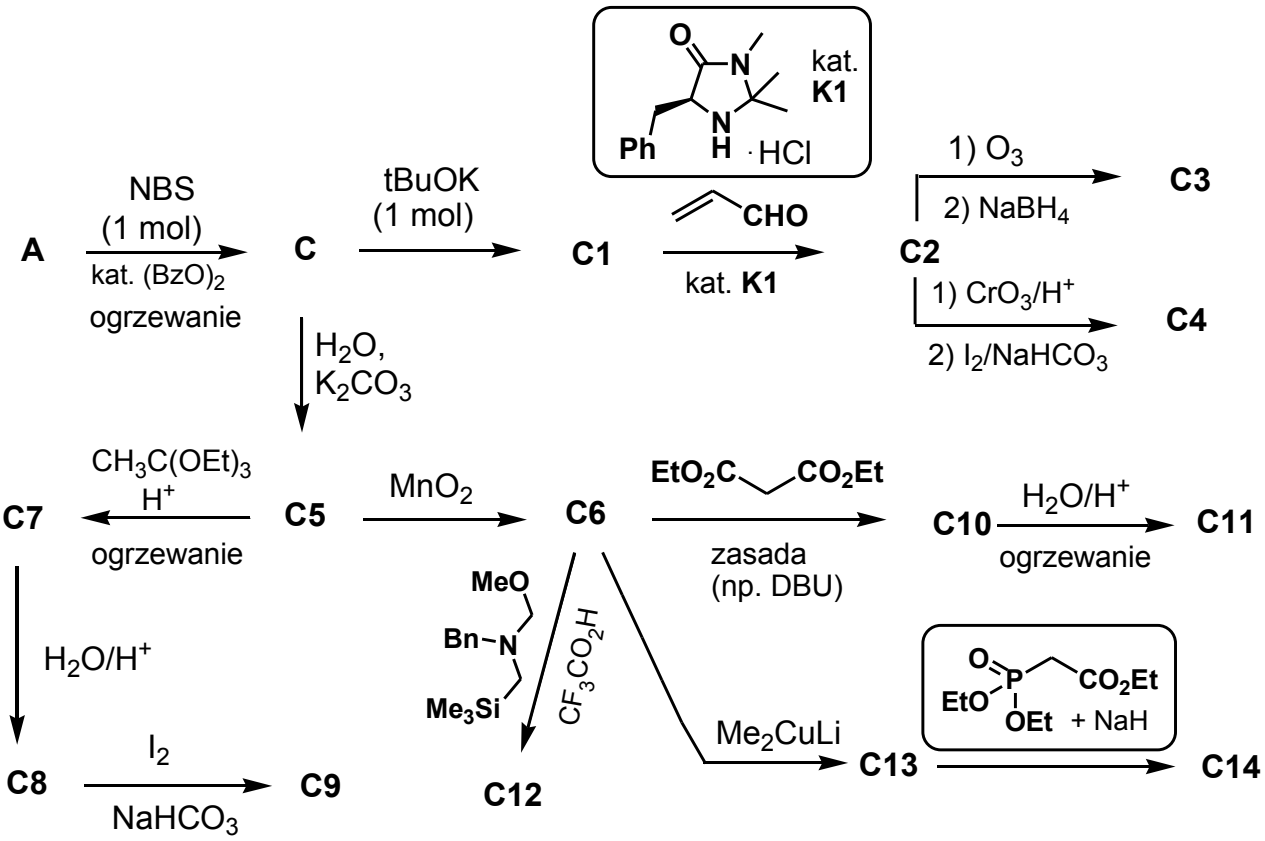
Wskazówki do schematu 1:

W widmie ^{13}C NMR związku **B1** widoczne są cztery sygnały. Związek **B4** zawiera 80,0% wagowych węgla. W reakcji prowadzącej do **B5** jednym z etapów jest przegrupowanie [3,3]-sigmatropowe oraz wydziela się amoniak.



Schemat 2

W przypadku związków **B11** i **B13** należy uwzględnić główny stereoizomer.



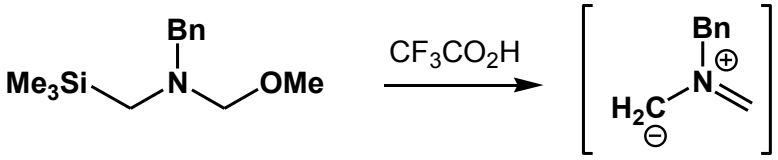
NBS – N-bromosukcynoimid; DBU - 1,8-diazabicyklo[5.4.0]undek-7-en

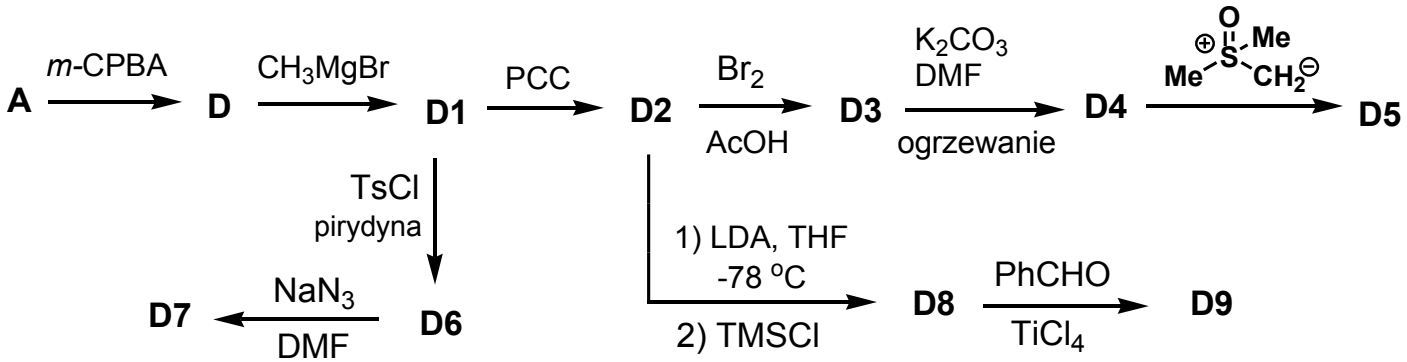
Schemat 3

<u>Wskazówki do schematu 3:</u>

W przypadku związków **C2, C3, C4, C9** i **C12** na schemacie 3 należy uwzględnić główne stereoizomery (konfiguracja względna).

Reakcja prowadząca do **C2** jest enancjoselektywna z uwagi na użyty homochiralny katalizator (nie jest wymagane, który enancjomer tworzy się w przewadze). Związek **C7** powstaje w wyniku przegrupowania [3,3]-sigmatropowego. Poniżej przedstawiono odczynnik generowany w reakcji prowadzącej do **C12**.



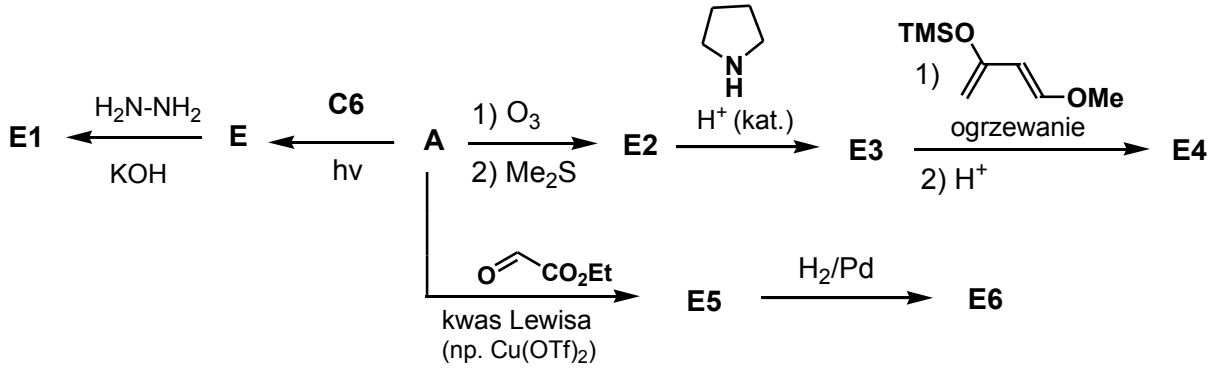


TsCl - chlorek p-toluenosulfonylu; TMSCl - chlorotrimetylosilan

Schemat 4

Wskazówki do schematu 4:

W przypadku związków **D1, D6** i **D7** na schemacie 4 należy uwzględnić główne stereoizomery.



Schemat 5

Wskazówki do schematu 5:

W widmie ^{13}C NMR głównego produktu **E1** widoczne są trzy sygnały, a jeden z pierścieni ma część podstawników w ułożeniu trans.

Związek **E4** zawiera dwa atomy tlenu wchodzące w skład grup karbonylowych.

Związek **E5** jest mieszaniną diastereoizomerów i jest produktem reakcji enowej, zaliczanej do grupy reakcji pericyklicznych.

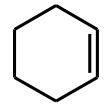
Związek **E6** zawiera jedno centrum stereogeniczne.

W przypadku związków **E, E1** i **E4** należy uwzględnić główne stereoizomery.

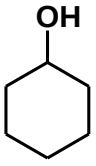
Polecenie:

Narysuj wzory strukturalne związków **A**, **B-B13**, **C-C14**, **D-D9**, **E-E6**

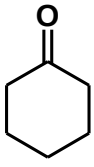
ROZWIĄZANIE ZADANIA B7



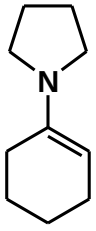
A



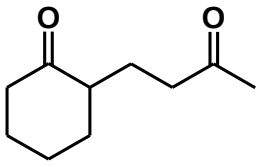
B



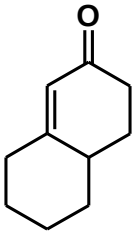
B1



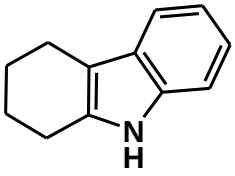
B2



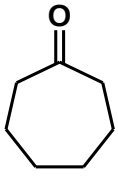
B3



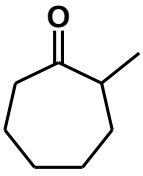
B4



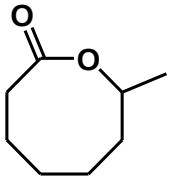
B5



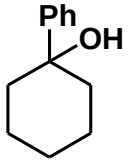
B6



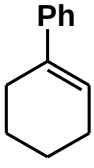
B7



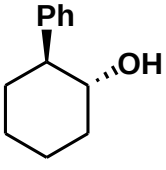
B8



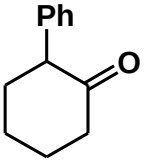
B9



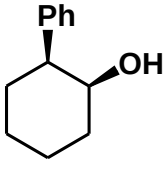
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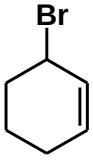
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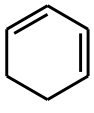
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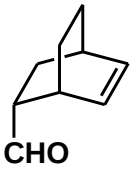
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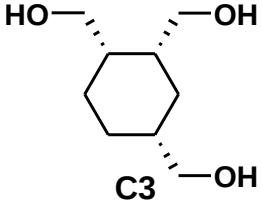
C



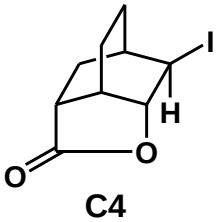
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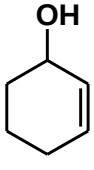
C2



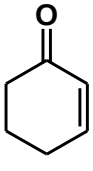
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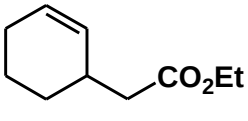
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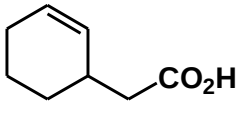
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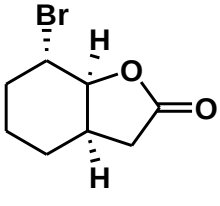
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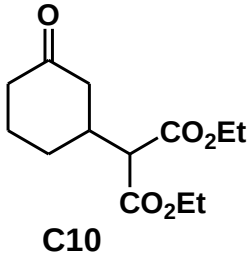
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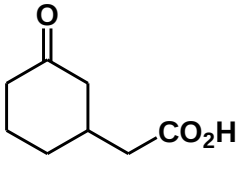
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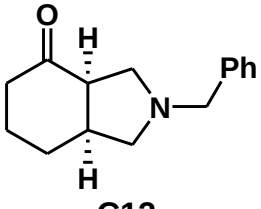
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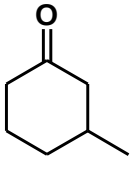
C10



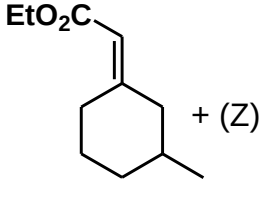
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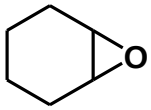
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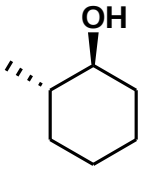
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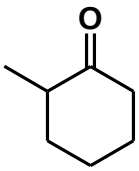
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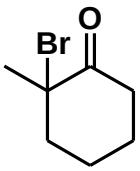
D



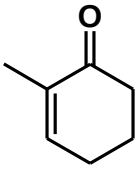
D1



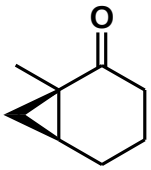
D2



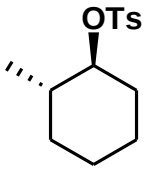
D3



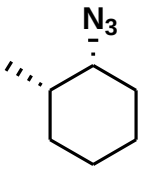
D4



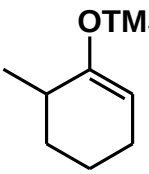
D5



D6

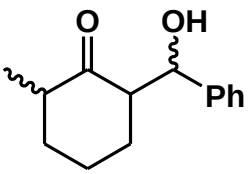


D7

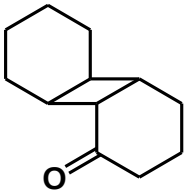


D8

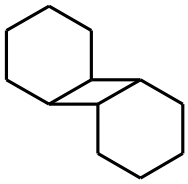
produkt kinetyczny



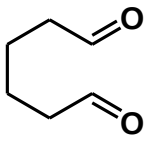
D9



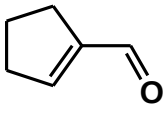
E



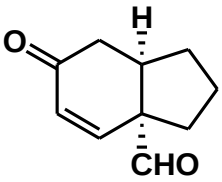
E1



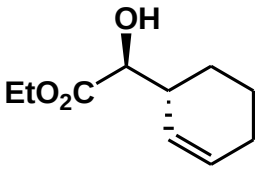
E2

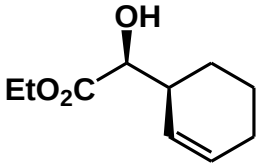


E3

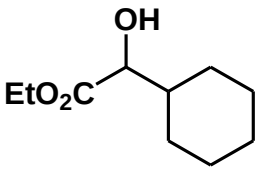


E4





E5 (diastereoizomery)



E6